

Scrolling through the past: How digital tools change the way we remember

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We are constantly documenting our lives with digital technologies. But how do these tools, from smartphone camera rolls to wearables and social media platforms, change what and how we remember? In this article, we explore the interplay between memories stored in our minds and the available technological devices.

Memories about our lives

Autobiographical memories are commonly defined as memories about our own lives (such as your birthday two years ago). These memories do not remain unchanged our whole lives, as our current goals and priorities can influence what we focus on, what we remember most clearly, or what we tell others about. We keep this information about the events of our lives as internal memories in our minds. In addition, we may have stored information about these events in external resources such as photo albums and diaries. This is nothing new: Humans have been using these kinds of external resources for centuries or even millennia. But with the digital landscape surrounding us, the data stored on our phones is denser, searchable, and accessible anytime and anywhere, offering unprecedented possibilities for remembering.

So, how can we analyze and understand the changes to autobiographical remembering in the digital age from a psychological point of view? One recent suggestion in this regard is provided by the **AMEDIA-model** (Autobiographical Memory in the Digital Age), which tries to break down the iterative process of autobiographical remembering, in which people combine their internal memories with information from external digital sources. The model outlines how we record digital information about our lives (**encoding**), change or reorganize it (**curation**), and later recall it (**retrieval**).

Encoding: More than just taking a picture

Encoding autobiographical information depends on what we pay attention to and which details we want to remember later. Choosing what to record is often intentional and active, but it doesn't end there. With digital tools constantly around us, many aspects of our lives are now captured automatically: by our phones, apps, or even by other people.

What's new and striking today are the tools that have no real analogue equivalent in the past. Social media platforms, for example, create vast archives of shared memories. As a result, the way we later remember an event crucially depends on the type of digital tools we used and the purpose we had in mind.

Curation: Editing our digital records

Externally recorded information can be changed, rearranged, or reorganized: a process that can be described as **curation**. This curation of our data may alter how we remember certain events in the future, especially when we delete particular information. Meanwhile, algorithms also curate our data by organizing it in meaningful ways. In the end, externally stored data is not static but formed and shaped by our interactions with it.

Retrieval: Reliving the past through our phones

Retrieval refers to the process of trying to remember something from your past. Here lies the most important interaction between internal memory and external recorded

information, as using external technical devices may influence what you remember and how you recall events from the past.

Having access to photos and other digital traces can enhance remembering or provide new insights when internal memory is vague, helping people compensate for details that may fade over time, while also shaping how past events are experienced and understood in the present.

Retrieval isn't just about pulling up data from the past. Browsing through your digital resources, especially on social media, can lead to reminiscing or even nostalgia, but can also lead to rumination when digital data confronts us with memories we would rather forget.

Looking ahead: Opportunities and risks of digital remembering

New questions arise for research and practice. In which contexts can these tools truly support remembering, and when might they change or even distort it?

Digitally capturing and later reviewing everyday experiences can improve older adults' ability to re-experience the past. The use of **Digital Life Stories** (i.e., digital photo albums enhanced with music, voice recordings, and memory prompts) for people with dementia has gained growing attention in clinical practice and research.

However, remembering always remains a reconstructive process: Photos, messages, and posts do not speak for themselves but need to be interpreted. Digital content can potentially be manipulated, and deepfakes and misleading online posts can contribute to false memories.

What this means for our memories

We now carry vast digital archives of our own lives in our pockets. Understanding what this means for remembering requires looking at what we record, how we curate these digital traces, and how we later use them to recall the past. Many open questions

remain, but one thing is already clear: how we document our lives also shapes how we remember and make sense of them. Reflecting on our own digital habits may be a first step toward using these tools more thoughtfully.

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